Caring for a Client with a Neurologic Disorder

Neuro Assessment

Baseline: compare with the child's usual behavior and caregiver report.

LOC first: alertness, orientation/response and pediatric GCS (eye, verbal, motor).

Check: pupil size/equality/reaction, limb movement/strength and sensation.

Trend: HR, BP, breathing pattern, temperature, oxygen saturation and pain.

Infants: head circumference, fontanel, sutures, feeding, cry and tone.

Urgent change: falling LOC, new weakness, unequal/nonreactive pupils or seizures. [1]



Diagnostic Tests

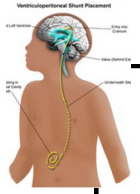
CT: rapid injury/bleeding assessment. **MRI:** detailed brain/spine anatomy.

Head ultrasound: infant ventricles through an open fontanel. **EEG:** seizure activity.

LP/CSF: cells, glucose, protein, Gram stain and culture. Obtain ordered blood cultures.

LP safety: suspected dangerous raised ICP/mass effect needs medical review first. Do not delay antibiotics.

Labs: glucose, electrolytes and CBC as indicated. Monitor after sedation/procedures. [2-5]



VP Shunt

Purpose: drains ventricular CSF to the peritoneum.

Assess: neuro checks, infant head growth, incision and abdomen.

After surgery: follow positioning/activity orders. Avoid pressure on the incision.

Report urgently: headache, vomiting, unusual sleepiness, bulging fontanel, fever, redness or abdominal pain.

Teach: follow-up and malfunction signs. Programmable valve: ask about checks after MRI. [5,6]

Medication Management

Seizures: rescue midazolam/diazepam per plan. Maintenance medicines as ordered.

Raised ICP: ordered hypertonic saline. Monitor sodium, fluids and neuro status.

Infection: IV antibiotics. Selected cases: dexamethasone.

Spasticity: baclofen or botulinum toxin when prescribed.

Safety: verify weight-based doses. Check breathing/sedation. Do not stop antiseizure drugs abruptly. [3,7,8,11]

Intracranial Pressure

VS?

Rising ICP: headache, vomiting, irritability and reduced alertness. Infants may have a bulging fontanel and high-pitched cry.

Late VS pattern (Cushing triad): hypertension with widening pulse pressure, bradycardia and irregular respirations. EMERGENCY.

Nursing: protect oxygenation/perfusion. Keep head midline, elevate about 30 degrees if ordered, and avoid neck/hip flexion or straining.

Monitor: serial neuro checks, fluid balance and ordered ICP/CPP targets. CPP = MAP - ICP. Targets vary by age/condition. [1,2,8]

Seizure NC

During: stay, call for help, time the seizure, clear hazards and protect the head. Turn to the side when safe.

Airway: observe breathing. Keep oxygen/suction ready. Check glucose when feasible.

Never: restrain movements or put anything in the mouth. No oral intake until fully alert.

Escalate: seizure at 5 minutes, repeated seizures without recovery, or breathing difficulty. Give prescribed rescue medicine per plan.

After: reassess airway/LOC, allow rest, document movements and recovery, and teach the seizure action plan. [7,9]

Neuro

Hydrocephalus



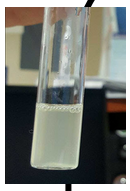
WHAT / WHY
CSF builds up in ventricles from blocked flow or poor absorption.

CLUES
Fast head growth, bulging fontanel, separated sutures, downward "sunset" eyes. Older child: headache, vomiting or drowsiness.

CARE
Trend head size and neuro status. Assess feeding, vomiting and skin.

TREAT / TEACH
VP shunt or selected endoscopic surgery. Keep follow-up. Return urgently for renewed symptoms or fever. [5,6]

Bacterial Meningitis



WHAT / WHY
Bacterial infection inflames the meninges. Risk of sepsis and raised ICP.

CLUES
Fever, headache, stiff neck, photophobia. Infant: poor feeding, irritability, bulging fontanel. Nonblanching rash is urgent.

CARE / TREAT
Prompt IV antibiotics. Cultures/LP when safe without treatment delay. Monitor neuro status, seizures and fluids.

PREVENT
Meningococcal/Hib: droplet + standard until 24 h effective antibiotics. Vaccines, indicated contact prophylaxis and hearing follow-up. [3,10]

Head Injury



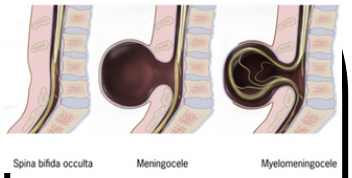
WHAT / WHY
Trauma causes concussion, bleeding or swelling. Hypoxia and hypotension worsen injury.

CLUES
Altered LOC, headache, vomiting, seizures or unequal pupils.

PRIORITIES
ABCs and cervical-spine protection if suspected injury. Serial GCS, pupils and vital signs. CT when indicated.

TREAT / TEACH
Manage ICP/bleeding as ordered. Escalate worsening drowsiness, repeated vomiting or new deficits. After concussion, gradual return to school/activity with clearance for sport. [2,8]

Myelomeningocele



WHAT / WHY
Neural tube defect with meninges and spinal cord/nerves in an open back sac.

CLUES
Leg weakness or sensory loss, bladder/bowel dysfunction. Hydrocephalus may occur.

BEFORE REPAIR
Prone or side-lying. Protect sac with a sterile, saline-moistened nonadherent dressing per order. Prevent drying/soiling. Use latex-free supplies.

AFTER / TEACH
Monitor infection, skin, head growth and movement. Follow the bladder/bowel program and catheterization plan. [12,13]

Cerebral Palsy



WHAT / WHY
Nonprogressive injury/abnormal development of the immature brain affects movement and posture.

CLUES
Delayed motor skills, abnormal tone, spasticity/scissoring or poor coordination. Symptoms can change with growth.

CARE
Support feeding/swallowing, communication, skin and mobility. Prevent contractures. Assess seizures.

TREAT / TEACH
Early PT/OT/speech therapy, braces and selected medicines/surgery. Support independence and family goals. [11]

Neuro concept maps: reference notes

Completed study notes for the original two-page worksheet. References checked October 12, 2026.

[1] Neurological assessment and deterioration - Royal Children's Hospital. Open reference

[2] Pediatric head injury - Royal Children's Hospital. Open reference

[3] Meningitis and encephalitis - Royal Children's Hospital. Open reference

[4] Lumbar puncture safety - Royal Children's Hospital. Open reference

[5] Hydrocephalus - Seattle Children's. Open reference

[6] Pediatric VP shunt surgery and care - Memorial Sloan Kettering. Open reference

[7] Seizures: acute management - Royal Children's Hospital. Open reference

[8] Pediatric severe traumatic brain injury management - Pediatric Critical Care Medicine. Open reference

[9] Seizure first aid - CDC. Open reference

[10] Meningococcal disease and infection control - CDC. Open reference

[11] Cerebral palsy: overview and treatment - CDC / NIH. Open reference

[12] Myelomeningocele: postnatal closure - UC Davis Children's Hospital. Open reference

[13] Spina bifida - NINDS. Open reference

Abbreviations: LOC = level of consciousness; GCS = Glasgow Coma Scale; CSF = cerebrospinal fluid; VP = ventriculoperitoneal; ICP = intracranial pressure; CPP = cerebral perfusion pressure; MAP = mean arterial pressure; NC = nursing care; PT/OT = physical/occupational therapy.

Medication selection, doses, positioning and monitoring follow the child's orders and local protocol. These concise entries are study notes, not an instructor-issued answer key. Original worksheet artwork and headings remain on pages 1-2.